B-Spline

Ör/ $P_0[1\,1]$, $P_1[2\,3]$, $P_2[4\,3]$, $P_3[3\,1]$ köşe noktaları veriliyor 2. dereceden B-spline eğrilerini çizelim.

$$P(t) = P_0 \cdot N_{0,2}(t) + P_1 \cdot N_{1,2}(t) + P_2 \cdot N_{2,2}(t) + P_3 N_{3,2}(t) \Rightarrow \sum_{i=0}^{n} P_i \cdot N_{i,k}(t) \quad (k=2)$$

$$N_{i,k}(t) = \frac{(t-x_i)\cdot N_{i,k-1}(t)}{x_{i+k-1}-x_i} + \frac{(x_{i+k}-t)\cdot N_{i+1,k-1}(t)}{x_{i+k}-x_{i+1}}$$

$$N_{i,1} = \begin{cases} 1 & \text{if } x_i \le t \le x_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

n = köşe sayısı − 1

$n = 4-1 \Rightarrow 3$

$k=2$ için

$t_{max} = n-k+2 = 3-2+2 = 3$

$n = 4-1 = 3$

$k = 2$ $\quad n-k+2 = 3-2+2 = 3$

$[0\,0\,1\,2\,3\,3]$
$x_0\ x_1\ x_2\ x_3\ x_4\ x_5$

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$$N_{0,2}(t) = \frac{(t-x_0)\cdot N_{0,1}(t)}{x_1-x_0} + \frac{(x_2-t)\cdot N_{1,1}(t)}{x_2-x_1}$$

$$N_{1,2}(t) = \frac{(t-x_1)\cdot N_{1,1}(t)}{x_2-x_1} + \frac{(x_3-t)\cdot N_{2,1}(t)}{x_3-x_2}$$

$$N_{2,2}(t) = \frac{(t-x_2)\cdot N_{2,1}(t)}{x_3-x_2} + \frac{(x_4-t)\cdot N_{3,1}(t)}{x_4-x_3}$$

$$N_{3,2}(t) = \frac{(t-x_3)\cdot N_{3,1}(t)}{x_4-x_3} + \frac{(x_5-t)\cdot N_{4,1}(t)}{x_5-x_4}$$

$t=0$ için

$$N_{0,2}(0) = \frac{(0-0)\cdot}{} + \frac{(1-0)\cdot N_{1,1}(t)}{1-0} = \frac{1\cdot 1}{1} = 1$$

$$N_{1,2}(0) = \frac{(0-0)\cdot}{} + \frac{(2-0)\cdot N_{2,1}(t)}{2-1} = 0$$

$$N_{2,2}(0) = \frac{(0-2)\cdot N_{2,1}(t)}{2-1} + \frac{(3-0)\cdot N_{3,1}(t)}{3-2} = 0$$

$$N_{3,2}(0) = \frac{(0-3)\cdot N_{3,1}(t)}{} + \frac{(3-0)\cdot N_{4,1}(t)}{} = 0$$

$$N_{1,1} = \begin{cases} 1 & x_1 \le 0 \le x_2 \\ & 0 \le 0 < 1 \ \checkmark \end{cases}$$

$$N_{2,1} = \begin{cases} 1 & x_2 \le 0 \le x_3 \\ 0 \checkmark & 1 \le 0 < 2 \ \times \end{cases}$$

$$N_{3,1} = \begin{cases} 1 & x_3 \le 0 \le x_4 \\ 0 \checkmark & 2 \le 0 < 3 \ \times \end{cases}$$

$$P(0) = P_0 \cdot N_{0,2}(t) + P_1 \cdot N_{1,2}(t) + P_2 \cdot N_{2,2}(t) + P_3 \cdot N_{3,2}(t)$$

$$= P_0 \cdot 1$$

$$P(0) = [1\ 1]$$

ÖC/ $P_0[1\ 1]$ $P_1[2\ 3]$ $P_2[4\ 5]$ $P_3[3\ 1]$ $k=4$ için ... $t=0,5$ için

$\sum_{i=0}^{n} P_i \cdot N_{i,k}(t)$ $\quad n = 4-1 = 3$ $\quad t_{max} = n-k+2$

$3-4+2 = 1$

$P_0 \cdot N_{0,4}(t) + P_1 \cdot N_{1,4}(t) + P_2 \cdot N_{2,4}(t) + P_3 \cdot N_{3,4}(t)$

$t_{max} = \overset{n=3}{3-k+2} \rightarrow k=4$ $\quad [0000\,1111]$

$x_0\ x_1\ x_2\ x_3\ x_4\ x_5\ x_6\ x_7$

$[0000\,1111]$

$$N_{i,k}(t) = \frac{(t-x_i)\cdot N_{i,k-1}(t)}{x_{i+k-1}-x_i} + \frac{(x_{i+k}-t)\cdot N_{i+1,k-1}(t)}{x_{i+k}-x_{i+1}}$$

$$N_{0,4}(t) = \frac{(t-x_0)\cdot N_{0,3}(t)}{x_3-x_0} + \frac{(x_4-t)\cdot N_{1,3}(t)}{x_4-x_1}$$

$$N_{2,4}(t) = \frac{(t-x_2)\cdot N_{2,3}(t)}{x_5-x_2} + \frac{(x_6-t)\cdot N_{3,3}(t)}{x_6-x_3} \rightarrow N_{1,4}(t) = \frac{(t-x_1)\cdot N_{1,3}(t)}{x_4-x_1} + \frac{(x_5-t)\cdot N_{2,3}(t)}{x_5-x_2}$$

$$N_{3,4}(t) = \frac{(t-x_3)\cdot N_{3,3}(t)}{x_6-x_3} + \frac{(x_7-t)\cdot N_{4,3}(t)}{x_7-x_4}$$

$N_{0,3} = \begin{cases} 1 & x_1 \le t < x_2 \\ 0 & \text{r.y.} \end{cases}$

$N_{0,4}(0,5) = \frac{(0,5-0)\cdot N_{0,3}(0,5)}{0} + \frac{(1-0,5)\cdot N_{1,3}(0,5)}{1-0} \rightarrow M_3 = \begin{cases} 1 & x_1 \le t \le x_2 \\ 0 & 0 \le 0,5 \le 0 \end{cases}$

$N_{1,4}(0,5) = \frac{(0,5-0)\cdot N_{1,3}(0,5)}{1-0} + \frac{(1-0,5)\cdot N_{2,3}(0,5)}{1-0}$ $\quad N_{2,3} = \begin{cases} 1 & x_2 \le t < x_3 \\ 0 & 0 \le 0,5 \le x_3 \end{cases}$

$N_{2,4}(0,5) = \frac{(0,5-0)\cdot N_{2,3}(0,5)}{1-0} + \frac{(1-0,5)\cdot N_{3,3}(0,5)}{1-0}$ $\quad N_{3,3} = \begin{cases} 1 & x_3 \le t < x_4 \\ 0 & 0 \le 0,5 \le 1 \end{cases}$ ✓

$N_{3,4}(0,5) = \frac{(0,5-0)\cdot N_{3,3}(0,5)}{1-0} + \frac{(1-0,5)\cdot N_{4,3}(t)}{0}$ $\quad = 0,5$ ✓ $\quad = 0,5$ ✓

$P(0,5) = P_0 \cdot N_{0,4}(0,5) + P_1 \cdot N_{1,4}(0,5) + P_2 \cdot N_{2,4}(0,5) + P_3 \cdot N_{3,4}(0,5)$

$= [4\ 3] \cdot 0,5 + [3\ 1] \cdot 0,5$

$= [2\ 1,5] + [1,5\ 0,5]$

$= [3,5\ 2]$ ✓